Juan Antonio Barragan Noguera

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Summary of Qualifications

Computer Science Ph.D. student with a strong background in robotics, computer vision, machine learning, and simulation for surgical applications. Experienced in developing real-time systems that integrate medical imaging and sensorized instrumentation to improve surgical performance. Additional experience developing virtual reality (VR) and augmented reality (AR) systems for surgical navigation, training, and data generation.

Education

Johns Hopkins University, Baltimore, MD Aug. 2021 – Present
Ph.D. student in Computer Science, advised by Dr. Peter Kazanzides

- Expected graduation: February 2027

Purdue University, West Lafayette, IN Aug. 2019 – July 2021
M.Sc.Eng. in Industrial Engineering, advised by Dr. Juan P. Wachs

Universidad Nacional de Colombia, Bogotá, Colombia Aug. 2013 – Dec. 2018
B.S. in Electronic Engineering

Professional Experience

Laboratory for Computational Sensing and Robotics (LCSR) Aug. 2021 – Present
Computer Science Ph.D. program, Johns Hopkins University Baltimore, MD

- Developed 3D reconstruction algorithms for surgical scenes based on structure-from-motion, stereo depth, and robot kinematics for augmented reality (AR) navigation. Work under review at Healthcare Technology Letters 2026.
- Developed vision-based tracking algorithms to correct misalignment of AR overlays of critical structures caused by camera motion or deformation of the surgical bed.
- Designed registration strategies between vision-based 3D models and cone-beam CT and ultrasound images.
- Implemented navigation systems for skull base surgery based on an online digital twin of the patient running in a C++ physics simulator.
- Developed a virtual reality (VR) interface for digital twin navigation and evaluated the system on mastoidectomy training tasks.
- Conducted user studies comparing visual, auditory, and haptic feedback, derived from signed distance fields to critical structures during mastoidectomy, to improve situational awareness. Published at IROS 2023; submitted to IEEE Robotics and Automation Letters 2026.

Institute of Image-Guided Surgery of Strasbourg Jan. 2025 – Nov. 2025
Research Fellowship with Dr. Nicolas Padoy Strasbourg, France

- Led a research project in collaboration with clinicians from Gemelli Hospital (Rome, Italy), developing AI systems that improve intraoperative assessment of cancer spread during diagnostic laparoscopy.
- Developed an AI-based registration strategy to correlate preoperative CT imaging with intraoperative laparoscopic video during cancer spread assessment.
- Fine-tuned Vision Transformer backbones using self-supervised learning on a dataset of diagnostic laparoscopy videos.
- Developed models for the downstream task of abdominal region identification to correlate video and CT on a region-by-region basis.

Robotics & Digital Solutions at Johnson & Johnson, Algorithms/AI team May 2022 – Aug. 2022
Machine Learning Summer Internship with Dr. Hedyeh Rafii-Tari Redwood City, CA

- Conceived and implemented machine learning skill analysis algorithms using kinematic data from a teleoperated robotic bronchoscopy system.
- Built a skill analysis annotation pipeline for clinical data.

Funding and Fellowships

Chateaubriand Fellowship: $17,500 June 2024
Next generation of surgical simulation environments for training and synthetic data generation

Awards and Honors

RSJ/KROS Distinguished Interdisciplinary Research Award Aug. 2021
30th IEEE International Conference on Robot & Human Interactive Communication

Best Presentation Award July 2021
Raven/dVRK research workshop at the Hamlyn Symposium 2021

Exemption from undergraduate tuition, Universidad Nacional de Colombia 2014 – 2017
Awarded to the 15 students with the highest GPAs in the Electronic Engineering Department

Technical Skills

General: Surgical robotics, 3D Computer vision, Vision foundation models, AR/VR/XR
Programming languages: Python, C, C++
3D modelling and imaging software: Blender, 3D slicer, ImFusion Suite
Operating systems: MacOs, Linux and Windows
Python Libraries: Pandas, NumPy, and Pytorch
Robotics: ROS1, ROS2

Teaching Experience

Computer Science Department, Johns Hopkins University Summers of 2023 and 2024
Summer course instructor Baltimore, MD

- Taught an 8-week undergraduate-level course on the fundamentals of programming in C and C++.

Languages

- Spanish: native speaker
- French: proficient speaker, level B2 (DELF certification, 80.5/100)
- English: proficient speaker (TOEFL certification, 108/120)

Selected Publications

J. A. Barragan, B. Burkhart, et al., "A Fast Framework for Metric 3D Reconstructions and Vision-Based Tracking for Surgical Robotic Platforms," Healthcare Technology Letters, 2026. Under review.

J. Wang, **J. A. Barragan**, H. Ishida, J. Guo, Y.-C. Ku and P. Kazanzides, "A Digital Twin for Telesurgery Under Intermittent Communication," 2025 International Symposium on Medical Robotics (ISMR), Atlanta, GA, USA, 2025, pp. 218–224, doi: 10.1109/ISMR67322.2025.11025988.

C. J. Allison, H. Zhou, A. Munawar, P. Kazanzides and **J. A. Barragan**, "FIRE-3DV: Framework-Independent Rendering Engine for 3D Graphics Using Vulkan," 2024 Eighth IEEE International Conference on Robotic Computing (IRC), Tokyo, Japan, 2024, pp. 104–111, doi: 10.1109/IRC63610.2024.00053.

J. A. Barragan, H. Ishida, A. Munawar and P. Kazanzides, "Improving the realism of robotic surgery simulation through injection of learning-based estimated errors," 2024 International Symposium on Medical Robotics (ISMR), Atlanta, GA, USA, 2024, pp. 1–7, doi: 10.1109/ISMR63436.2024.10585672.

J. A. Barragan, J. Zhang, H. Zhou, A. Munawar and P. Kazanzides, "Realistic Data Generation for 6D Pose Estimation of Surgical Instruments," 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, 2024, pp. 13347–13353, doi: 10.1109/ICRA57147.2024.10611638.

H. Ishida, **J. A. Barragan**, et al., "Improving Surgical Situational Awareness with Signed Distance Field: A Pilot Study in Virtual Reality," 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Detroit, MI, USA, 2023, pp. 8474–8479, doi: 10.1109/IROS55552.2023.10342004.

J. A. Barragan, J. Yang, D. Yu, et al., "A neurotechnological aid for semi-autonomous suction in robotic-assisted surgery," Scientific Reports, vol. 12, Art. no. 4504, 2022, doi: 10.1038/s41598-022-08063-w.

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